

Intelligence Radar

Phase 2-4 Implementation Plan

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A rigorous, file-level implementation plan for transforming the Strategic Intelligence Radar from Phase 1 (orbital visualization + data hook) into a fully integrated intelligence system with signal diffing, scheduled research, prompt-level integration, health score contribution, and a professional World View experience.

Architecture Principles

- Zero new database tables -- all state in existing entities/edges tables + synthesis_data JSONB.
- Enhancement-not-replacement -- upgrade existing prompts and types; zero additional LLM calls for core features.
- Existing pattern compliance -- DashboardCard wrapping, SSE streaming, D3 SVG rendering, hook-based state management.
- Backward compatibility -- all new fields optional; existing dashboards render unchanged.
- Single atomic writes -- no clobber bugs; finalSynthData accumulation pattern throughout.

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Phase 2: Signal Intelligence + Interaction Depth

Phase 2 transforms the radar from a static visualization into a living intelligence feed. Signal diffing detects what changed between research runs. The connection analyzer gains deep mode linking external entities to specific strategy actions. The change alert module surfaces intelligence signals alongside structural changes.

2.1 Signal Diffing Engine

`src/lib/intelligence/compute-signals.ts`

A pure function that compares two ResearchSnapshot objects and produces IntelligenceSignal[] arrays. Called after every research run, signals are persisted in `synthesis_data.intelligence_radar.signals`.

Function Signature

```
computeSignals(oldSnapshot: ResearchSnapshot | null, newSnapshot: ResearchSnapshot,
currentEntities: Entity[], bridgeEdges: Edge[]): IntelligenceSignal[]
```

Diffing Logic

Signal Type (Severity)	Detection Logic
new_entity (HIGH)	Entity ID in new snapshot but not old. Severity HIGH if authority >= moderate AND has bridge to internal.
entity_removed (MEDIUM)	Entity ID in old snapshot but not new. Indicates landscape shift -- previously relevant context dropped.
authority_change (HIGH)	Same entity ID, different authority_level. HIGH severity: low->high (external validation gained). MEDIUM: high->low (source credibility dropped).
confidence_shift (LOW)	Same entity, confidence delta > 0.2. Indicates research found stronger/weaker evidence.
new_bridge (HIGH)	Edge in bridge set connects to entity not previously bridged. This is a novel connection between external intel and your system.
new_source (MEDIUM)	Entity's provenance.citation_urls grew. New web sources discovered for existing knowledge.

Integration Points

- Called from `research/route.ts` post-processing (after entities/edges stored, before response).
- Old snapshot read from `synthesis_data.intelligence_radar.last_snapshot`.
- New snapshot built from freshly-written entities: { entity_ids, authority_map, confidence_map, timestamp }.
- Signals merged with existing: keep last 50, deduplicate by entity_id + type, newest wins.
- Both snapshot and signals written to `synthesis_data.intelligence_radar` in a single atomic update.

CRITICAL: The research route must re-read `synthesis_data` fresh before writing (the strategy-refresh pattern at line 370) to avoid clobbering data written by other concurrent processes.

2.2 Research Snapshot Persistence

`src/app/api/pipeline/research/route.ts` (MODIFY lines 343-398)

After external entities and edges are stored in the database, and before the response is returned, inject a snapshot write and signal computation step.

Implementation Steps

1. After edge inserts (line ~380), query all external entities for this space: `db.from('entities').select('entity_id, authority_level, confidence').eq('space_id', spaceId).eq('knowledge_layer', 'external').`
2. Build ResearchSnapshot: `{ entity_ids: [...], authority_map: {...}, confidence_map: {...}, timestamp: new Date().toISOString() }`.
3. Read current synthesis_data fresh: `db.from('spaces').select('synthesis_data').eq('id', spaceId).single().`
4. Extract old snapshot: `synthData.intelligence_radar?.last_snapshot ?? null.`
5. Call `computeSignals(oldSnapshot, newSnapshot, entities, bridgeEdges).`
6. Merge signals: `[...existingSignals, ...newSignals].slice(0, 50)` with dedup.
7. Write back: `{ ...freshSynthData, intelligence_radar: { ...existingRadar, last_snapshot: newSnapshot, signals: mergedSignals, updated_at: now } }`.

2.3 Connection Analyzer Deep Mode

[src/lib/hooks/use-intelligence-radar.ts](#) (MODIFY `connectionAnalysis` function)

Enhance the existing `connectionAnalysis` function with multi-hop traversal and strategy-perspective scoring.

Enhancement	Implementation
Multi-hop bridges	Currently traces 1-hop bridges. Add 2-hop: external -> bridge entity -> internal. This catches indirect connections through bridge-layer entities.
Perspective scoring	For each connection, compute a <code>relevance_score</code> (0-1) based on: (a) edge confidence, (b) whether internal entity is leverage/risk/bottleneck (+0.2), (c) whether internal entity appears in active strategy perspective (+0.3).
Temporal relevance	If internal entity has <code>temporal_validity</code> with <code>decay_rate</code> , factor freshness into relevance. Stale connections score lower.
Worth-considering links	Cross-reference <code>worth_considering</code> items. If a WC item's <code>connected_entities</code> overlap with this external entity's bridges, surface it as a 'framework insight' link.

2.4 DashboardCard Fullscreen Pass-through

[src/components/dashboard/dashboard-card.tsx](#) (MODIFY)

The `DashboardCard` component manages fullscreen state internally but does not expose it to children. The Intelligence Radar needs to switch between compact and world-view layouts based on fullscreen state.

Changes Required

- Add optional prop: `renderContent?: (isFullscreen: boolean) => React.ReactNode.`
- When `renderContent` is provided, call it with current fullscreen state instead of rendering children directly.
- When `renderContent` is not provided, fall back to standard `{children}` rendering (backward compatible).
- This allows `IntelligenceRadarModule` to render `compactContent` when `!isFullscreen` and `fullscreenContent` (the World View) when `isFullscreen`.
- Alternative: Use React Context -- create `FullscreenContext` that `DashboardCard` provides, children consume via `useContext`. More elegant but higher coupling.

RECOMMENDATION: Use the `renderContent` callback pattern. It's explicit, type-safe, zero coupling, and only the Intelligence Radar module needs it. Other modules continue using `{children}` unchanged.

2.5 Change Alert Module Integration

[src/components/dashboard/modules/change-alert-module.tsx \(MODIFY\)](#)

Surface high-severity intelligence signals in the existing change alert system. Currently shows synthesis structural changes. Extend to include external landscape shifts.

- Add prop: `intelligenceSignals?: IntelligenceSignal[]`.
- In the change summary section (line ~114), append signal summary: 'N new external signals detected (K high-severity)'.
- Add expandable section below synthesis delta: list high-severity signals with entity name, category icon, description.
- Add 'View in Radar' link that scrolls to / opens the Intelligence Radar module.
- Trigger visibility: show alert when `signals.filter(s => s.severity === 'high').length > 0` AND signals detected_at within last 24h.

2.6 Node Detail External Enhancement

[src/components/graph/node-detail.tsx \(MODIFY\)](#)

When the user clicks an external entity in the main graph (not just the radar), the node detail slide-out should show intelligence-specific context.

- Detect `knowledge_layer === 'external'` and render enhanced provenance section.
- Show: category badge (with icon), authority badge, source URLs as clickable chips, relevance text.
- Show: 'Bridge connections' section listing all internal entities this connects to (reuse `ConnectionPanel` pattern).
- Show: 'Strategy relevance' section if this entity appears in any strategy perspective's `supporting_entities` array.
- Add 'Open in Radar' button that sets `selectedEntityId` in the Intelligence Radar hook (requires callback prop from parent).

Phase 3: Scheduled Research + Prompt Integration

Phase 3 makes the intelligence system proactive. Research runs on schedule. Prompts are upgraded to weight external evidence. Quality scoring and health metrics account for intelligence coverage. The system moves from 'show what we have' to 'actively seek what we need.'

3.1 Orchestrate researchOnly Mode

[src/app/api/orchestrate/route.ts \(MODIFY\)](#)

The orchestrate endpoint currently runs the full pipeline (decompose -> structure -> synthesize). Add a researchOnly flag that skips entity decomposition and only runs the domain expert research agent.

Implementation

1. Accept researchOnly: boolean in request body (line ~19).
2. When researchOnly=true, skip phases 1-5 (decomposition, structuring, dimension classification, reconciliation, quality).
3. Fetch existing entities/edges from DB (they already exist from prior analysis).
4. Run only Agent 7 (domain expert) with existing entities as context.
5. Run bridge discovery against existing internal entities.
6. Store results in entities/edges tables (external entities, bridge edges).
7. Run snapshot + signal computation (Phase 2.2).
8. Emit SSE events: 'phase: researching' -> 'phase: bridge_discovery' -> 'complete'.
9. Return: { success, entitiesCreated, edgesCreated, signalsDetected }

Cost: ~\$0.02-0.08 per research run (1 LLM call + 0-12 web searches at \$0.01 each). Latency: 5-15s depending on depth. No synthesis cost.

3.2 Focused/Targeted Research

[src/lib/prompts/domain-expert.ts \(MODIFY\)](#)

Add ability to focus research on specific categories or search terms, enabling infrastructure proposal 'Research this' buttons and scheduled research with focus areas.

Prompt Modifications

- Add optional focus_areas: string[] parameter to getDomainExpertPrompt().
- When focus_areas provided, prepend to system prompt: 'RESEARCH FOCUS: Prioritize the following areas: {focus_areas.join(', ')}. Allocate 70% of your entities and all web searches to these areas. Cover remaining categories lightly.'
- Add optional skip_categories: ExternalCategory[] parameter. When provided: 'SKIP these categories entirely: {skip}. Do not generate entities in these categories.'
- Modify web search triggering: When focus_areas includes a category, ALWAYS search for that category regardless of confidence threshold.
- Add intent: string parameter for infrastructure proposal research: 'RESEARCH INTENT: The user is evaluating whether to build/adopt: {intent}. Focus on: tools, platforms, and workflows related to this intent. Include pricing, alternatives, integration patterns, and known failure modes.'

Data Flow

```
ResearchSchedule.focus_areas -> orchestrate endpoint -> runDomainExpert(scope, input, domains, { focus_areas }) -> getDomainExpertPrompt(scope, domains, { focus_areas, skip_categories }) -> research execution.
```

3.3 Research Schedule UI

[src/components/intelligence/schedule-panel.tsx \(NEW\)](#)

A settings panel accessible from the Intelligence Radar module header. Allows users to configure research cadence, depth, and focus areas.

UI Components

- Cadence selector: Radio group -- Manual / Daily / Weekly / Biweekly / Monthly. Default: Weekly.
- Depth selector: Segmented control -- Light (3 searches) / Standard (6) / Deep (12). Default: Standard.
- Focus areas: Tag input with autocomplete from existing ExternalCategory values + freetext. Up to 5 tags.
- Toggle: 'Auto-research enabled' switch with next_run_at preview: 'Next run: April 16, 2026'.
- Research history: Last 5 runs as compact list: date, depth, entities found, signals generated.
- Manual trigger: 'Research Now' button with depth override dropdown.
- Renders inside the World View right panel OR as a popover from the compact header.

Persistence

All schedule state persisted via POST /api/pipeline/research-schedule (already built in Phase 1). Stored in synthesis_data.intelligence_radar.schedule. The useIntelligenceRadar hook's auto-trigger mechanism (checking next_run_at on mount) provides client-side scheduling without server cron.

3.4 Domain Expert Prompt Upgrade

[src/lib/prompts/domain-expert.ts \(MODIFY\)](#)

Beyond focus areas (3.2), upgrade the domain expert to produce richer output for the radar.

Upgrade	Implementation
Relevance scoring	Add relevance_score: 1-10 to each external entity. Prompt instruction: 'Score each entity 1-10 for relevance to the user's specific situation, not generic domain importance. A score of 10 means this entity directly impacts a decision the user faces.'
Temporal freshness flag	Add temporal_freshness: 'current' 'aging' 'historical' to entity output. Web-searched data defaults to 'current'. Training data older than 2 years: 'historical'. Between: 'aging'.
Connection hints	Add likely_connects_to: string[] to each entity (internal entity names, not IDs, that this likely relates to). This seeds bridge discovery with stronger priors.
Risk signal flag	Add is_risk_signal: boolean. True if this entity represents a threat, competitive move, regulatory change, or negative trend that the user should monitor.

3.5 Synthesis Quality External Dimension

[src/lib/validation/synthesis-quality.ts \(MODIFY\)](#)

Add a 6th scoring dimension: External Coverage. Currently 5 dimensions sum to 100%. Rebalance weights to accommodate.

New Dimension: External Coverage (weight: 10%)

Scoring Factor	Points
External entity count	+15 if external_entities >= 5, +10 if >= 3, +5 if >= 1
Cross-context insights	+20 if insights >= 3, +15 if >= 2, +10 if >= 1

Scoring Factor	Points
Worth-considering with external source	+10 per item with source_origin === 'external_research' (max 30)
Bridge coverage	+15 if bridges connect to >= 50% of leverage/risk entities
Authority quality	+10 if authority_distribution.high >= 30% of external entities
Overconfidence penalty	-20 if synthesis references external data but has ZERO contradictions or challenges from external sources

Weight Rebalancing

Dimension	Current Weight	New Weight
Depth	25%	22%
Specificity	20%	18%
Utility Grounding	15%	14%
Evidence Quality	20%	18%
Actionability	20%	18%
External Coverage	0%	10%

3.6 Health Score External Coverage

[src/lib/twin/compute-twin-state.ts](#) (MODIFY)

Add external intelligence coverage as a factor in the Digital Twin health score. Currently health_score ranges 0-100 with no external component.

New Health Score Component: Intelligence Coverage (up to +8 pts)

- +3 if external_entities.length >= 3 (basic landscape awareness).
- +2 if bridge_edges.length >= 2 (connected to internal system).
- +2 if cross_context_insights.length >= 1 (insights spanning both).
- +1 if research_provenance.searches_performed >= 1 (web-verified data).
- -3 if external_entities.length === 0 AND synthesis has 5+ leverage/risk points (blind spot penalty -- complex system with no external context)

Net effect: A fully-researched space can score up to 8 points higher. A complex space with zero research gets a -3 penalty, surfacing the blind spot in the Digital Twin.

3.7 Goal Recommendations External Weighting

[src/lib/prompts/goal-recommendations.ts](#) (MODIFY)

Currently, goal recommendations rank findings but don't weight them by external validation. Add explicit instructions for external evidence weighting.

Prompt Addition (append to ranking rules)

EXTERNAL EVIDENCE WEIGHTING:\n- If an external entity (X-prefixed) VALIDATES a leverage point that supports this goal, boost that recommendation by +1 rank position.\n- If external data CHALLENGES an assumption underlying a recommendation, FLAG it with a warning note and lower confidence.\n- If a cross_context_insight specifically mentions the goal's target metric or

entity, cite it in the recommendation's reasoning.\n- If worth_considering items of type 'precedent' exist, reference them as supporting evidence.

Phase 4: Full Integration + Professional Polish

Phase 4 elevates the Intelligence Radar to a professional-grade experience. The World View becomes the primary intelligence interface. Infrastructure proposals trigger targeted research. Bridges gain approval workflows. The system becomes self-improving through cross-module event flow.

4.1 World View Full-Screen Experience

```
src/components/intelligence/world-view.tsx (NEW)
```

Extract the fullscreen layout from intelligence-radar-module.tsx into a dedicated component. This becomes the primary intelligence workspace with rich interactivity.

Layout Architecture

Three-panel layout with responsive proportions:

Panel	Contents
Left Panel (15%)	Entity inventory grouped by ring, then by category. Collapsible groups. Click to select. Drag to reorder priority. Badge counts per group. Search/filter at top.
Center Panel (55%)	Full orbital graph with smooth physics. Ring labels clickable as filters. Background pulse animation on high-severity signals. Minimap in bottom-right corner for orientation. Category legend overlay (toggleable).
Right Panel (30%)	Context-sensitive: Default = summary + signals + schedule. Entity selected = ConnectionPanel deep mode + provenance + strategy links. Category filter = grouped entity list with authority bars. Signal selected = full signal detail with affected entities highlighted on graph.

Interaction Enhancements

- Double-click external node: Open provenance detail with source URLs, authority history, temporal freshness.
- Right-click external node: Context menu with 'Research deeper', 'Find connections', 'Mark as key intel', 'Dismiss'.
- Drag from external node to core: Manual bridge creation (prompts for relationship type).
- Keyboard shortcuts: R = research now, F = toggle filters, 1-4 = ring filters, Esc = deselect.
- Animated bridge paths: Bridges pulse with a traveling dot animation showing information flow direction.
- Signal indicators: High-severity signal entities get a pulsing red ring overlay.

4.2 Infrastructure Proposal Research Buttons

```
src/components/dashboard/modules/strategy-recommendation-module.tsx (MODIFY lines 495-540)
```

Each infrastructure proposal card (tool, app, dashboard, workflow) gains a 'Research this' button that triggers targeted research focused on that proposal's domain.

Implementation

- Add prop: onResearchProposal?: (proposal: InfrastructureProposal) => void to StrategyRecommendationModule.
- Render button in the proposal card's preview area (line ~517): 'Research this' with Globe2 icon.
- On click: Call orchestrate with researchOnly=true, focus_areas=[proposal.name], intent=proposal.description.
- Show inline loading state: spinner replacing button text, 'Researching {proposal.name}...'.
- On completion: router.refresh() to reload radar with new external entities.
- Default recommendation: When a strategy is first generated and has infrastructure proposals, show a one-time callout: 'New infrastructure recommendations detected. Research external tools and frameworks to validate

these proposals?'

4.3 Bridge Approval Workflow

Currently bridges are created by LLM reasoning but have no approval lifecycle. Add a lightweight approval flow so users can validate, reject, or strengthen connections.

Database Changes

```
Add migration: ALTER TABLE edges ADD COLUMN bridge_status TEXT DEFAULT 'auto' CHECK (bridge_status IN ('auto', 'confirmed', 'rejected', 'pending'));
```

UI Integration

- In ConnectionPanel: Show 'Confirm' / 'Reject' buttons on each bridge connection.
- Confirmed bridges: Solid gold line in orbital graph (currently dashed). Weight +0.2 in connection strength.
- Rejected bridges: Hidden from radar. Edge marked bridge_status='rejected' in DB. Still stored for audit.
- Pending bridges: Pulsing dashed line. Shown with '?' badge. Prompt user to review.
- In World View right panel: 'Pending Reviews' section at top when pending bridges exist.
- API endpoint: PATCH /api/edges/[id]/bridge-status with { status: 'confirmed' | 'rejected' }.

4.4 Objective Detection External Enrichment

```
src/lib/prompts/objective-detection.ts (MODIFY)
```

The objective detection prompt already references external context (Phase 3 integration at lines 108-120). Enhance to produce richer external_evidence on detected objectives and to generate intelligence-informed objectives that wouldn't exist without external data.

- Add to Phase 1 (Backward Chain): 'When tracing causal chains, if an external entity (X-prefixed) appears in the chain, annotate the convergence point with external_evidence showing HOW external data strengthens the chain.'
- Add new Phase 5: INTELLIGENCE-DRIVEN OBJECTIVES: 'Identify 1-2 objectives that ONLY become visible through external context. Examples: competitive positioning objectives visible only from competitor data, risk mitigation objectives from external risk patterns, opportunity objectives from cross-domain analogies.'
- Enrich external_evidence type: Add strength: 'strong' | 'moderate' | 'weak' and entity_ids: string[] referencing specific external entities that support this evidence.
- Add to propellant identification: 'If an external framework or resource could accelerate the propellant, reference it: propellant.external_accelerator = { entity_id, mechanism }'.

4.5 Cross-Module Event System

```
src/lib/events/intelligence-events.ts (NEW)
```

A lightweight pub/sub system for cross-module communication. Enables the Intelligence Radar to notify other modules when external context changes, and other modules to request intelligence on specific entities.

Event	Payload	Flow
intelligence:signals_detected	IntelligenceSignal[]	Radar -> Change Alert, Objective Command Center
intelligence:research_complete	{ depth, entities_added, bridges_added }	Radar -> Strategy Module, Synthesis Modules
intelligence:entity_selected	{ entity_id, ring }	Radar -> Node Detail, Graph Module

Event	Payload	Flow
strategy:research_requested	{ intent, focus_areas }	Strategy Module -> Radar
objectives:external_evidence_needed	{ objective_key }	Objective CC -> Radar

Implementation: React Context + useReducer pattern. EventBus provider at SpaceLayout level. Modules subscribe via useIntelligenceEvent(eventName, handler). Events are fire-and-forget, synchronous within the React render cycle. No persistence -- events are ephemeral UI coordination.

4.6 Research Changelog + History

[src/components/intelligence/research-history.tsx \(NEW\)](#)

Track and display research history so users can see how their external landscape evolved over time. Stored in synthesis_data.intelligence_radar.history (array, last 20 entries).

History Entry Structure

```
{ timestamp, depth, trigger: 'manual' | 'scheduled' | 'proposal', entities_added, entities_removed, bridges_created, signals: IntelligenceSignal[], focus_areas, searches_performed, cost_estimate }
```

UI

- Timeline view in World View right panel: vertical list with date, depth badge, entity delta (+N/-M), signal count.
- Click entry to see details: full signal list, entities added/removed, bridges created.
- Sparkline chart at top: entity count over time (last 20 research runs).
- Export: 'Download intelligence report' button generating a summary of all external entities, bridges, and insights.

4.7 Category Depth Layering

[src/components/intelligence/depth-layers.tsx \(NEW\)](#)

The 'layers of logic + depth + type of correlation' visualization. Groups external entities into nested layers that show increasing analytical depth.

Layer	Categories	Description
Layer 1: Raw Data	data_point, resource	Direct facts: numbers, benchmarks, tools. Lowest inference.
Layer 2: Patterns	pattern, risk_pattern, competitor	Observed patterns: competitive moves, known risks, structural regularities. Medium inference.
Layer 3: Frameworks	framework, analogy	Interpretive frames: methodologies, cross-domain analogies. Highest inference.

- Visual: Three concentric zones in the orbital graph, replacing the current ring labels with depth labels when 'Depth View' toggle is active.
- Correlation lines between layers: Show how a data_point validates a pattern which informs a framework. These are the existing edges between external entities, but re-colored by depth transition (green = data->pattern, purple = pattern->framework).
- Filter interaction: Click a layer to isolate it. Double-click to see only cross-layer connections.
- This view answers: 'What raw evidence supports which patterns, and which frameworks interpret those patterns?'

Dependency Graph + Build Order

Tasks are ordered by dependency. Items at the same level can be built in parallel. Critical path is highlighted.

Phase 2 Build Order (Week 1-2)

Task	Name	Depends On	Scheduling	Estimate
2.1	Signal Diffing Engine	None	CRITICAL PATH	4h
2.2	Research Snapshot Persistence	2.1	CRITICAL PATH	2h
2.4	DashboardCard Fullscreen Pass-through	None	Parallel with 2.1	1h
2.3	Connection Analyzer Deep Mode	None	Parallel with 2.1	3h
2.5	Change Alert Integration	2.1, 2.2	After 2.2	2h
2.6	Node Detail Enhancement	2.3	After 2.3	2h

Phase 3 Build Order (Week 3-4)

Task	Name	Depends On	Scheduling	Estimate
3.1	Orchestrate researchOnly Mode	2.2	CRITICAL PATH	3h
3.2	Focused/Targeted Research	3.1	CRITICAL PATH	3h
3.4	Domain Expert Prompt Upgrade	3.2	After 3.2	2h
3.3	Research Schedule UI	3.1	Parallel with 3.2	3h
3.5	Synthesis Quality External Dimension	None	Parallel with 3.1	2h
3.6	Health Score External Coverage	None	Parallel with 3.1	1h
3.7	Goal Recommendations External Weighting	None	Parallel with 3.1	1h

Phase 4 Build Order (Week 5-7)

Task	Name	Depends On	Scheduling	Estimate
4.1	World View Full-Screen	2.4, 2.3	CRITICAL PATH	6h
4.2	Infrastructure Research Buttons	3.1, 3.2	After Phase 3	2h
4.3	Bridge Approval Workflow	None	Parallel with 4.1	4h
4.4	Objective Detection Enrichment	None	Parallel with 4.1	2h
4.5	Cross-Module Event System	4.1, 4.2	After 4.1	3h

Task	Name	Depends On	Scheduling	Estimate
4.6	Research Changelog + History	2.2, 3.3	After Phase 3	3h
4.7	Category Depth Layering	4.1	After 4.1	4h

Total Estimates

Phase	Hours	Summary
Phase 2	14 hours	Signal diffing, snapshots, interaction depth, UI integration
Phase 3	15 hours	Scheduled research, prompt upgrades, quality/health scoring
Phase 4	24 hours	World View polish, bridge workflow, events, depth layers
TOTAL	53 hours	7 weeks at ~8 hours/week

Risk Assessment

Risk	Severity	Description	Mitigation
Research cost accumulation	HIGH	Scheduled research at 'deep' depth (12 searches) costs ~\$0.14/run. Daily schedule = ~\$4.20/month. Weekly = ~\$0.60/month.	Default to 'standard' depth (6 searches, ~\$0.08/run). Show cost estimate in schedule UI. Cap at 'standard' for daily cadence. Allow 'deep' only for weekly+.
Snapshot storage growth	LOW	Each snapshot stores entity_ids + authority_map + confidence_map. At 50 external entities, ~2KB per snapshot.	Only keep last_snapshot (not history of snapshots). History entries store only delta counts, not full snapshots. JSONB compression handles the rest.
Signal noise	MEDIUM	Frequent research may generate many low-value signals (minor confidence shifts, training data fluctuations).	Only surface signals with severity 'high' or 'medium' in the Change Alert module. Low-severity signals visible only in the Intelligence Radar signal feed. Auto-expire signals older than 7 days.
Orchestrate endpoint complexity	MEDIUM	Adding researchOnly flag creates a conditional code path that may diverge from the main pipeline over time.	Extract research logic into a standalone function: runResearchPhase(spaceId, depth, focusAreas). Called by both orchestrate (full pipeline) and researchOnly mode. Single source of truth.
Bridge approval fatigue	LOW	Users may ignore pending bridge approvals, leaving bridges in 'pending' state indefinitely.	Default bridge_status to 'auto' (current behavior). Only show approval UI in the World View, not in the main dashboard. Auto-promote bridges with confidence >= 0.8 to 'confirmed' after 7 days if not rejected.
DashboardCard fullscreen coupling	LOW	Adding renderContent callback may create API coupling between DashboardCard and the Intelligence Radar.	Keep renderContent optional. Only Intelligence Radar uses it. All other modules continue with {children}. Document the pattern for future modules.
External entity ID collisions	MEDIUM	Multiple research runs may create duplicate external entities with different X-prefixed IDs for the same real-world concept.	Before inserting new external entities, run a name-similarity check against existing external entities in the space. If > 85% name match, update the existing entity instead of creating a new one. Log the merge in research history.

Complete File Manifest

Every file created or modified across all three phases.

New Files (8)

File Path	Phase	Purpose
src/lib/intelligence/compute-signals.ts	2.1	Signal diffing engine

File Path	Phase	Purpose
src/components/intelligence/schedule-panel.tsx	3.3	Research schedule UI
src/components/intelligence/world-view.tsx	4.1	Full-screen World View layout
src/components/intelligence/research-history.tsx	4.6	Research changelog + timeline
src/components/intelligence/depth-layers.tsx	4.7	Category depth layering view
src/lib/events/intelligence-events.ts	4.5	Cross-module event bus
supabase/migrations/YYYYMMDD_bridge_status.sql	4.3	Bridge status column migration
src/app/api/edges/[id]/bridge-status/route.ts	4.3	Bridge approval API

Modified Files (14)

File Path	Phase	Changes
src/app/api/pipeline/research/route.ts	2.2	Add snapshot write + signal computation post-processing
src/lib/hooks/use-intelligence-radar.ts	2.3	Multi-hop connection analysis, deep mode
src/components/dashboard/dashboard-card.tsx	2.4	Add renderContent callback for fullscreen detection
src/components/dashboard/modules/change-alert-module.tsx	2.5	Surface intelligence signals
src/components/graph/node-detail.tsx	2.6	Enhanced external entity provenance display
src/app/api/orchestrate/route.ts	3.1	Add researchOnly flag + conditional pipeline flow
src/lib/prompts/domain-expert.ts	3.2, 3.4	Focus areas, relevance scoring, temporal freshness
src/lib/validation/synthesis-quality.ts	3.5	Add External Coverage scoring dimension
src/lib/twin/compute-twin-state.ts	3.6	Add intelligence coverage to health score
src/lib/prompts/goal-recommendations.ts	3.7	External evidence weighting rules
src/components/dashboard/modules/strategy-recommendation-module.tsx	4.2	Infrastructure proposal research buttons
src/lib/prompts/objective-detection.ts	4.4	Intelligence-driven objectives, enriched external_evidence
src/components/dashboard/modules/intelligence-radar-module.tsx	4.1	Wire World View via renderContent
src/components/layout/space-layout.tsx	4.5	EventBus provider at layout level